United States Patent Application Publication	20250282142
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	ISHIZAWA; Taku et al.

LIQUID EJECTION APPARATUS AND CONTROL METHOD FOR LIQUID EJECTION APPARATUS

Abstract

A liquid ejection apparatus includes a liquid ejection unit configured to eject liquid and a control unit. The control unit executes calculation processing of calculating an evaporation amount of waste liquid by which the waste liquid evaporates from a waste liquid container provided with an atmosphere opening hole, and estimation processing of estimating a contained amount of the waste liquid contained in the waste liquid container, based on an additional amount of the waste liquid by which the liquid ejected by the liquid ejection unit is added to the waste liquid container as the waste liquid and a result in the calculation processing.

Inventors: **ISHIZAWA; Taku (MATSUMOTO-SHI, JP), KOMATSU; Yuta (SHIOJIRI-SHI, JP), SAWADAISHI; Makoto (SHIOJIRI-SHI, JP)**

Applicant: **SEIKO EPSON CORPORATION** (TOKYO, JP)

Family ID: **1000008531487**

Appl. No.: **19/076073**

Filed: **March 11, 2025**

Foreign Application Priority Data

JP	2024-036751	Mar. 11, 2024

Publication Classification

Int. Cl.: **B41J2/17** (20060101); **B41J2/175** (20060101)

U.S. Cl.:

CPC **B41J2/1728** (20240501); **B41J2/17566** (20130101); B41J2002/17573 (20130101)

Background/Summary

[0001] The present application is based on, and claims priority from JP Application Serial Number 2024-036751, filed Mar. 11, 2024, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to a liquid ejection apparatus and a control method for the liquid ejection apparatus.

2. Related Art

[0003] For example, Japanese Patent Application Laid-Open No. 2003-145793 discloses a liquid ejection apparatus to which a waste liquid container can be attached, and liquid ejected from a liquid ejection unit is contained as waste liquid in the waste liquid container. In such a liquid ejection apparatus, a liquid amount detection unit can detect that the contained amount of the waste liquid has reached the upper limit contained amount.

[0004] Furthermore, as such a liquid ejection apparatus, there is also known an apparatus that calculates the contained amount of the waste liquid contained in the waste liquid container based on the additional amount of the waste liquid added to the waste liquid container and issues a notification related to the contained amount of the waste liquid contained in the waste liquid container. For example, a notification for the replacement of the waste liquid container can be issued in advance, with a notification issued indicating that the contained amount of the waste liquid contained in the waste liquid container has reached a prescribed contained amount less than the upper limit contained amount.

[0005] The waste liquid container includes a container provided with an atmosphere opening hole in order to increase the contained amount of the waste liquid. When such a waste liquid container is used, the evaporation of the waste liquid can be promoted through the atmosphere opening hole.

[0006] However, with such a liquid ejection apparatus in which the contained amount of the waste liquid contained in the waste liquid container is estimated, there is a concern that the estimation accuracy of the contained amount of the waste liquid may be compromised due to a decrease in the contained amount of the waste liquid depending on the evaporation amount of the waste liquid. In view of the above, there has been a demand for improvement in the function related to the contained amount of the waste liquid contained in the waste liquid container.

SUMMARY

[0007] A liquid ejection apparatus to satisfy the demand described above includes a liquid ejection unit configured to eject liquid, a liquid amount detection unit configured to detect, in a waste liquid container configured to contain liquid ejected by the liquid ejection unit as waste liquid, that a contained amount of waste liquid contained in the waste liquid container reaches a threshold, and a control unit, the control unit being configured to execute calculation processing of calculating an evaporation amount of waste liquid when waste liquid contained in the waste liquid container evaporates from the waste liquid container provided with an atmosphere opening hole, and estimation processing of estimating a contained amount of waste liquid contained in the waste liquid container, based on an additional amount of waste liquid when liquid ejected by the liquid ejection unit is added to the waste liquid container as waste liquid and a calculation result in the calculation processing.

[0008] A control method for a liquid ejection apparatus to satisfy the demand described above is a control method for a liquid ejection apparatus configured to perform control related to a contained amount of waste liquid that is liquid ejected by a liquid ejection unit and contained in a waste liquid container as waste liquid, the control method including executing determination processing of determining that a contained amount of waste liquid contained in the waste liquid container reaches a threshold, based on a detection result by a liquid amount detection unit, calculation processing of calculating an evaporation amount of waste liquid when waste liquid contained in the waste liquid container evaporates from the waste liquid container provided with an atmosphere opening hole, and estimation processing of estimating a contained amount of waste liquid contained in the waste liquid container, based on an additional amount of waste liquid when liquid ejected by the liquid ejection unit is added to the waste liquid container as waste liquid and a calculating result in the calculation processing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0009] FIG. **1** is a schematic view illustrating a liquid ejection apparatus according to a first embodiment.
[0010] FIG. **2** is a perspective view illustrating a waste liquid container according to the first embodiment.
[0011] FIG. **3** is a perspective view illustrating the waste liquid container according to the first embodiment.
[0012] FIG. **4** is a flowchart illustrating waste liquid control processing according to the first embodiment.
[0013] FIG. **5** is a timing chart illustrating the relationship between a waste liquid contained amount and time according to the first embodiment.
[0014] FIG. **6** is a perspective view illustrating a waste liquid container according to a second embodiment.
[0015] FIG. **7** is a flowchart illustrating waste liquid control processing of the second embodiment.
DESCRIPTION OF EMBODIMENTS
First Embodiment
[0016] An embodiment of a liquid ejection apparatus and a control method for the liquid ejection apparatus will be described below with reference to the drawings.
Configuration of Liquid Ejection Apparatus **10**
[0017] A liquid ejection apparatus **10** is configured to perform printing by ejecting liquid onto a medium **90**, as illustrated in FIG. **1**. The liquid ejection apparatus **10** may be an ink-jet printer that performs printing by ejecting ink as an example of the liquid to the medium **90**. The medium **90** may be, for example, paper, cloth, vinyl, a plastic component, a metal component, or the like.
[0018] The liquid ejection apparatus **10** includes a liquid ejection unit **11**. The liquid ejection unit **11** is configured to eject liquid onto the medium **90**. The liquid ejection unit **11** includes a liquid ejection head **12**. The liquid ejection head **12** ejects the liquid to the medium **90**. The liquid ejection head **12** is configured to eject the liquid contained in a liquid container **50** described below.
[0019] The liquid ejection head **12** includes a plurality of nozzles **13**. The liquid ejection head **12** performs printing by ejecting liquid onto the medium **90** from each of the plurality of nozzles **13**. The liquid ejection head **12** may be of a serial type or a line type. The liquid ejection head **12** of a serial type performs printing by ejecting liquid onto the medium **90** while moving. The liquid ejection head **12** of a line type is provided in an elongated form in a width direction of the medium **90**, and performs printing by ejecting liquid onto the medium **90**.
[0020] The liquid ejection apparatus **10** may include a conveyance unit **14**. The conveyance unit **14** is configured to convey the medium **90**. The conveyance unit **14** conveys the medium **90** to the liquid ejection head **12** from a medium container not illustrated. The conveyance unit **14** conveys the medium **90** after the printing by the liquid ejection head **12** to the outside of the liquid ejection apparatus **10**. The conveyance unit **14** may include a roller.

[0021] The liquid ejection apparatus **10** may include an attachment portion **20**. The liquid container **50** is attachable to and detachable from the attachment portion **20**. A plurality of the liquid containers **50** corresponding to the respective colors of the liquid may be attachable to the attachment portion **20**. The liquid container **50** contains liquid to be supplied to the liquid ejection head **12**. The liquid container **50** may be an ink cartridge.
[0022] The liquid ejection apparatus **10** may include a supply flow path **22**. The liquid ejection apparatus **10** may include a plurality of the supply flow paths **22** corresponding to the respective colors of the liquid. The supply flow path **22** is a flow path for supplying liquid from the liquid container **50** to the liquid ejection head **12** when the liquid container **50** is attached to the attachment portion **20**. When the liquid container **50** is attached to the attachment portion **20**, an upstream end of the supply flow path **22** is coupled to the liquid container **50**. A downstream end of the supply flow path **22** is coupled to the liquid ejection head **12**.
[0023] The supply flow path **22** may enable the liquid contained in the liquid container **50** to be led out when a supply needle **23** is inserted into the liquid container **50** attached to the attachment portion **20**. The supply flow path **22** may supply the liquid by means of hydraulic head. The liquid ejection apparatus **10** may include a pump (not illustrated) for supplying liquid from the liquid container **50** to the liquid ejection head **12**.
[0024] The liquid ejection apparatus **10** may include a cap **30**. The cap **30** can receive liquid discharged from the liquid ejection head **12**. The cap **30** may be movable between a contact position illustrated in FIG. **1** and a separated position not illustrated. The contact position is a position at which the cap **30** is in contact with the liquid ejection head **12**. The separated position is a position at which the cap **30** is separated from the liquid ejection head **12**.
[0025] The cap **30** located at the contact position forms a closed space **31** to which each of the plurality of nozzles **13** opens. Maintenance performed with the cap **30** forming the closed space **31** surrounding the plurality of nozzles **13** is also referred to as capping.
[0026] The cap **30** at the separated position may receive liquid ejected from the liquid ejection head **12**. Maintenance performed with the liquid ejection head **12** ejecting liquid is also referred to as flushing.
[0027] The liquid ejection apparatus **10** may include a cleaning unit **32**. The cleaning unit **32** can perform cleaning. Cleaning is maintenance performed with liquid forcibly discharged from each of the plurality of nozzles **13** via the closed space **31**. Thus, the cleaning unit **32** can forcibly discharge the liquid from the liquid ejection head **12**.
[0028] The cleaning unit **32** may include a waste liquid flow path **33** and a suction pump **34**. The waste liquid flow path **33** is a flow path for discharging liquid as waste liquid from the cap **30** to a waste liquid container **51** described below, when the waste liquid container **51** is attached to a waste liquid attachment portion **35**. An upstream end of the waste liquid flow path **33** is coupled to the cap **30**. When the waste liquid container **51** is attached to the waste liquid attachment portion **35**, a downstream end of the waste liquid flow path **33** is coupled to the waste liquid container **51**. The waste liquid flow path **33** may enable the liquid as the waste liquid to be led out from the waste liquid container **51** when a discharge needle **36** is inserted into the waste liquid container **51** attached to the waste liquid attachment portion **35**.
[0029] The suction pump **34** can reduce the pressure in the space inside the cap **30** through the waste liquid flow path **33**. The cleaning unit **32** performs cleaning by driving the suction pump **34** with the cap **30** located at the contact position. In other words, the cleaning unit **32** depressurizes the closed space **31**. Accordingly, the liquid is discharged from each of the plurality of nozzles **13**, and the liquid is supplied from the supply flow path **22** to the liquid ejection head **12**.
[0030] The cleaning unit **32** may discharge the liquid in the cap **30** by driving the suction pump **34** with the cap **30** located at the separated position. Maintenance performed with the liquid in the cap **30** sent to the waste liquid container **51** described below is also referred to as idle suction.
[0031] The liquid ejection apparatus **10** may include the waste liquid attachment portion **35**. The

waste liquid container **51** is attachable to and detachable from the waste liquid attachment portion **35**. The waste liquid container **51** can contain liquid as waste liquid discharged from the liquid ejection head **12** by the cleaning unit **32**.

[0032] The liquid ejection apparatus **10** includes a liquid amount detection unit **37**. For the waste liquid container **51**, the liquid amount detection unit **37** detects that the contained amount of the waste liquid contained in the waste liquid container **51** has reached a threshold. The liquid amount detection unit **37** detects that the contained amount of the waste liquid contained in the waste liquid container **51** has reached an upper limit contained amount as the threshold. The liquid amount detection unit **37** may be sensors. The upper limit contained amount is a threshold of the contained amount that is an upper limit of the amount of the waste liquid that can be contained in the waste liquid container **51**.

[0033] Specifically, the liquid amount detection unit **37** may be an optical sensor. The liquid amount detection unit **37** includes a light emitting portion AI and a light receiving portion **37**B. The light emitting portion **37**A emits light toward a reflection portion **55** of the waste liquid container **51** described below. The light receiving portion **37**B receives light reflected from the reflection portion **55**. The liquid amount detection unit **37** detects that the contained amount of the waste liquid contained in the waste liquid container **51** has reached the upper limit contained amount based on the light received by the light receiving portion **37**B.

[0034] The liquid ejection apparatus **10** may include a container type detection unit **38**. The container type detection unit **38** can detect that the waste liquid container **51** is or is not attached to the waste liquid attachment portion **35**. The container type detection unit **38** can detect the type of the waste liquid container **51** attached to the waste liquid attachment portion **35**. The container type detection unit **38** may be sensors.

[0035] The liquid ejection apparatus **10** includes a control unit **40** and an input unit **41**. The liquid ejection apparatus **10** may include a notification unit **42**. The control unit **40** comprehensively controls driving of each mechanism in the liquid ejection apparatus **10** and controls various operations performed in the liquid ejection apparatus **10**.

[0036] The control unit **40** can be configured as a circuit including a: one or more processors that execute various processes in accordance with a computer program, β: one or more dedicated hardware circuits that execute at least some of the various processes, or y: a combination thereof. The hardware circuit is, for example, an application specific integrated circuit. The processor includes a CPU and a memory such as a RAM and a ROM, and the memory holds program codes or commands configured to cause the CPU to execute processing. The memory, i.e., a computer readable medium, includes any type of readable mediums that are accessible by general-purpose or dedicated computers.

[0037] The input unit **41** enables input of a predetermined instruction. The input unit **41** enables input of various instructions related to the liquid ejection apparatus **10**. The input unit **41** is configured as an operation unit that can be operated by a user. Alternatively, the input unit **41** may be configured to receive a predetermined instruction input on a terminal device (not illustrated). In this case, the input unit **41** may be configured separately from the control unit **40** or may be configured to be included in the control unit **40**.

[0038] The notification unit **42** issues a notification related to the liquid ejection apparatus **10**. The notification unit **42** may be at least one of a display unit capable of displaying an image, a sound output unit capable of outputting sound, and a light emitting portion capable of emitting light in a predetermined light emitting mode. The notification unit **42** may not be provided in the liquid ejection apparatus **10** and may be provided in a terminal device (not illustrated). In this case, the control unit **40** may output a notification instruction to a terminal device (not illustrated), and the notification unit **42** may issue a notification based on the notification instruction.

[0039] In the present embodiment, the control unit **40** controls the liquid ejection unit **11**, the conveyance unit **14**, the cap **30**, and the cleaning unit **32**. Based on detection signals from the liquid amount detection unit **37** and the container type detection unit **38**, the control unit **40** can recognize the detection result. The control unit **40** performs control based on an instruction input using the input unit **41**. The control unit **40** causes the notification unit **42** to issue a notification related to the liquid ejection apparatus **10**.

Configuration of Waste Liquid Container **51**

[0040] As illustrated in FIG. **2**, the waste liquid container **51** is configured to contain the liquid ejected by the liquid ejection unit **11** as waste liquid. The waste liquid container **51** is replaceable with respect to the liquid ejection apparatus **10**. The liquid ejection apparatus **10** may include the waste liquid container **51**.

[0041] The waste liquid container **51** includes a waste liquid container case **52** and a waste liquid container cover **53**. The waste liquid container case **52** is configured to contain waste liquid. The waste liquid container cover **53** is configured to cover the waste liquid container case **52** from above.

[0042] The waste liquid container case **52** includes an injection hole **54**. The injection hole **54** is configured to enable injection of the liquid ejected by the liquid ejection unit **11** into the waste liquid container **51** as waste liquid. When the discharge needle **36** is inserted into the injection hole **54**, the liquid as the waste liquid can be injected into the waste liquid container **51**.

[0043] The waste liquid container case **52** includes the reflection portion **55**. The reflection portion **55** is exposed from the waste liquid container case **52**. The reflection portion **55** is provided at a position to face the liquid amount detection unit **37**. The reflection portion **55** is a member for detecting that the waste liquid is or is not stored in a first storage portion **61** described below. The waste liquid stored in the first storage portion **61** changes a light reflection state of the reflection portion **55**. The reflection portion **55** may be a prism.

[0044] The waste liquid container case **52** includes an atmosphere opening hole **56**. Thus, the waste liquid container **51** includes the atmosphere opening hole **56**. The atmosphere opening hole **56** may be provided on the outer bottom surface side of the waste liquid container case **52**.

[0045] The atmosphere opening hole **56** communicates with the inside of the waste liquid container **51**. The atmosphere opening hole **56** is a hole through which the inside of the waste liquid container case **52** opens to the atmosphere. The atmosphere opening hole **56** is provided to promote evaporation of the waste liquid contained in the waste liquid container **51**. That is, the atmosphere opening hole **56** enables the evaporation of the waste liquid contained in the waste liquid container **51**. The atmosphere opening hole **56** reduces the contained amount of the waste liquid contained in the waste liquid container **51** through the evaporation of the waste liquid. As a result, the amount of waste liquid that can be contained in the waste liquid container **51** increases.

[0046] As illustrated in FIG. **3**, the waste liquid container case **52** includes a waste liquid containing portion **60**, the first storage portion **61**, and an atmosphere opening portion **63**. Thus, the waste liquid container **51** includes the waste liquid containing portion **60**, the first storage portion **61**, and the atmosphere opening portion **63**.

[0047] The waste liquid containing portion **60** can contain the waste liquid injected from the injection hole **54**. In other words, the waste liquid containing portion **60** can contain the liquid ejected from the liquid ejection unit **11** as the waste liquid. The waste liquid containing portion **60** includes a waste liquid containing chamber that contains the waste liquid in the waste liquid container **51**.

[0048] The waste liquid container case **52** includes an absorber **57**. The absorber **57** is disposed in the waste liquid containing portion **60**. In the drawing, a part of the absorber **57** is illustrated. The absorber **57** is a material that absorbs the waste liquid. The absorber **57** absorbs the waste liquid contained in the waste liquid containing portion **60**.

[0049] The waste liquid container case **52** includes a first partition wall **64**. The first partition wall **64** is provided between the waste liquid containing portion **60** and the first storage portion **61**. The first partition wall **64** provides a partition between the waste liquid containing portion **60** and the first storage portion **61**. The first partition wall **64** is provided between the waste liquid containing portion **60** and the atmosphere opening portion **63**. The first partition wall **64** provides a partition between the waste liquid containing portion **60** and the atmosphere opening portion **63**.

[0050] The first partition wall **64** is lower than an outer wall **52**A of the waste liquid container case **52**. Therefore, a gap is provided between the first partition wall **64** and the waste liquid container cover **53**. Accordingly, the waste liquid containing portion **60** and the first storage portion **61** communicate with the atmosphere opening portion **63**. Thus, the evaporation of the waste liquid contained in the waste liquid containing portion **60** and the first storage portion **61** is promoted through the atmosphere opening hole **56**.

[0051] The first partition wall **64** includes a first communication portion **65**. The first communication portion **65** is provided so as to establish communication between the waste liquid container **51** and the first storage portion **61**. The first communication portion **65** is provided at a position higher than an inner bottom surface **52**B of the waste liquid container case **52**. The first communication portion **65** may have a recessed shape.

[0052] The first storage portion **61** is provided in a flow path through which the waste liquid containing portion **60** and the atmosphere opening hole **56** communicate. The first storage portion **61** can store the waste liquid. Specifically, the first storage portion **61** includes a first storage chamber capable of storing the waste liquid. The reflection portion **55** is provided in the first storage portion **61**. The liquid amount detection unit **37** detects that the waste liquid of a first storage amount is contained in the first storage portion **61**, by receiving the light reflected by the reflection portion **55**. The first storage amount is an upper limit of the amount that can be contained in the first storage portion **61**, but may be less than the upper limit amount. The contained amount of the waste liquid contained in the waste liquid containing portion **60** and the first storage portion **61** corresponds to the upper limit contained amount of the waste liquid contained in the waste liquid container **51**.

[0053] The waste liquid container case **52** includes a second partition wall **66**. The second partition wall **66** is provided between the first storage portion **61** and the atmosphere opening hole **56**. The second partition wall **66** provides a partition between the first storage portion **61** and the atmosphere opening portion **63**.

[0054] The second partition wall **66** is lower than the outer wall **52**A of the waste liquid container case **52**. Therefore, a gap is provided between the second partition wall **66** and the waste liquid container cover **53**. Accordingly, the first storage portion **61** communicates with the atmosphere opening portion **63**. Thus, the evaporation of the waste liquid contained in the first storage portion **61** is promoted through the atmosphere opening hole **56**.

[0055] The atmosphere opening portion **63** is provided with the atmosphere opening hole **56**. The atmosphere opening portion **63** includes a water-repellent ventilation film **58**. The atmosphere opening hole **56** is provided with the water-repellent ventilation film **58**. The water-repellent ventilation film **58** is a film that has water repellency but allows gas to pass therethrough.

[0056] By using the water-repellent ventilation film **58**, the waste liquid contained in the waste liquid container **51** can be efficiently evaporated. By using the water-repellent ventilation film **58**, it is possible to prevent the waste liquid from leaking to the outside of the waste liquid container **51** through the atmosphere opening hole **56** when a certain amount of waste liquid enters the atmosphere opening portion **63**.

[0057] In this manner, the liquid ejected from the liquid ejection unit **11** is injected into the waste liquid container **51** through the injection hole **54**. The waste liquid injected into the waste liquid container **51** is contained in the waste liquid containing portion **60**. The waste liquid contained in the waste liquid containing portion **60** is absorbed by the absorber **57**.

[0058] The waste liquid failed to be absorbed by the absorber **57** in the waste liquid containing portion **60** enters the first storage portion **61** via the first communication portion **65**. The liquid amount detection unit **37** can detect that the waste liquid is or is not stored in the first storage portion **61**, via the reflection portion **55**. Specifically, the liquid amount detection unit **37** can detect that the waste liquid of the upper limit contained amount is or is not contained in the waste liquid container **51** via the reflection portion **55**.

[0059] If the waste liquid is further injected into the waste liquid container **51** after the waste liquid of the upper limit contained amount has been contained in the waste liquid containing portion **60** and the first storage portion **61**, the waste liquid overflowed from the waste liquid containing portion **60** and the first storage portion **61** enters the atmosphere opening portion **63**. Therefore, when the contained amount of the waste liquid contained in the waste liquid container **51** reaches the upper limit contained amount, the waste liquid container **51** needs to be replaced.

Waste Liquid Control Processing

[0060] Waste liquid control processing will be described with reference to FIG. **4**. The waste liquid control processing is processing executed by the control unit **40** at a predetermined cycle. The waste liquid control processing is processing of performing control related to the contained amount of the waste liquid contained in the waste liquid container **51**.

[0061] As illustrated in FIG. **4**, in step S**10**, the control unit **40** determines whether a waste liquid condition is satisfied. The waste liquid condition is satisfied when the maintenance is performed with the waste liquid led out to the waste liquid container **51**. As a specific example, the waste liquid condition may be satisfied when cleaning is performed. Upon determining that the waste liquid condition is not satisfied, the control unit **40** terminates the waste liquid control processing. When the control unit **40** determines that the waste liquid condition is satisfied, the processing proceeds to step S**11**.

[0062] In step S**11**, the control unit **40** executes evaporation amount calculation processing. In this processing, the control unit **40** reads, from a memory, the elapsed time after the previous satisfaction of the waste liquid condition. The control unit **40** reads the evaporation rate from the memory. The evaporation rate indicates the evaporation amount per unit time. The control unit **40** calculates the evaporation amount of the waste liquid based on the elapsed time and the evaporation rate. Specifically, the control unit **40** calculates the product of the elapsed time and the evaporation rate as the evaporation amount of the waste liquid.

[0063] Thus, the control unit **40** calculates the evaporation amount of the waste liquid by which the waste liquid contained in the waste liquid container **51** evaporates from the waste liquid container **51** provided with the atmosphere opening hole **56**. Such evaporation amount calculation processing corresponds to an example of calculation processing.

[0064] In step S**12**, the control unit **40** executes estimation processing. In this processing, the control unit **40** reads, from the memory, the contained amount of the waste liquid estimated at the point of the previous satisfaction of the waste liquid condition. The control unit **40** subtracts the evaporation amount of the waste liquid from the previous waste liquid contained amount. Thus, the control unit **40** estimates the current waste liquid contained amount not including the additional amount of the waste liquid. In this way, the control unit **40** estimates the contained amount of the waste liquid contained in the waste liquid container **51** based on the calculation result. Hereinafter, the contained amount of the waste liquid estimated by the control unit **40** is referred to as an estimated contained amount.

[0065] In step S**13**, the control unit **40** determines whether the estimated contained amount estimated in response to the current satisfaction of the waste liquid condition is less than the estimated lower limit amount. The control unit **40** calculates the product of a cumulative contained amount and a lower limit coefficient as the estimated lower limit amount. The cumulative contained amount is a cumulative amount of the waste liquid added to the waste liquid container **51** after the waste liquid container **51** has been attached to the waste liquid attachment portion **35**. The estimated lower limit amount is a lower limit amount of the waste liquid, of the cumulative contained amount, estimated to remain without evaporating. The lower limit coefficient is a coefficient indicating a lower limit ratio of the waste liquid not evaporating.

[0066] The waste liquid contains ink, water, and a solvent at a predetermined ratio. When the waste liquid evaporates, water contained in the waste liquid evaporates, but the ink and the solvent do not evaporate at room temperature. Therefore, the lower limit coefficient is determined so as to correspond to the ratio of components other than water contained in the waste liquid. The lower limit coefficient may be 0.6 for example.
[0067] When the control unit **40** determines that the estimated contained amount is equal to or more than the estimated lower limit amount, the processing proceeds to step S**15**. When the control unit **40** determines that the estimated contained amount is less than the estimated lower limit amount, the processing proceeds to step S**14**.
[0068] In step S**14**, the control unit **40** executes estimated contained amount correction processing. In this processing, the control unit **40** corrects the estimated lower limit amount as the estimated contained amount. The control unit **40** stores the time when the waste liquid condition is satisfied and the estimated contained amount in the memory.
[0069] In step S**15**, the control unit **40** executes cumulative contained amount addition processing. In this processing, the control unit **40** calculates the additional amount of the waste liquid added when the waste liquid condition is satisfied. The additional amount of the waste liquid is an amount corresponding to the maintenance satisfying the waste liquid condition. As a specific example, when the cleaning is performed once, the additional amount of the waste liquid is an amount corresponding to the one cleaning. The control unit **40** records a result obtained by adding the calculated additional amount of the waste liquid to the cumulative contained amount in the memory as the current cumulative contained amount.
[0070] As described above, the control unit **40** calculates the additional amount of the waste liquid by which the liquid ejected by the liquid ejection unit **11** is added to the waste liquid container **51** as the waste liquid. This estimated contained amount addition processing corresponds to an example of the calculation processing.
[0071] In step S**16**, the control unit **40** executes estimated contained amount addition processing. In this processing, the control unit **40** records a result obtained by adding the calculated additional amount of the waste liquid to the estimated contained amount in the memory as the current estimated contained amount. In this way, the control unit **40** estimates the contained amount of the waste liquid contained in the waste liquid container **51** based on the calculation result and the additional amount of the waste liquid. The estimation processing and the estimated contained amount addition processing correspond to an example of estimation processing.
[0072] In step S**17**, the control unit **40** determines whether an error condition is satisfied. The error condition is satisfied when the estimated contained amount is equal to or more than an upper limit determination amount but the amount is detected by the liquid amount detection unit **37** to be not equal to or more than the upper limit contained amount. The upper limit determination amount is an amount with which the estimated contained amount can be assumed to have reached the upper limit contained amount. The upper limit determination amount may be equal to or more than the upper limit contained amount or may be less than the upper limit contained amount.
[0073] The error condition is satisfied when the estimated contained amount is equal to or less than a lower limit determination amount but the amount detected by the liquid amount detection unit **37** is equal to or more than the upper limit contained amount. The lower limit determination amount is an amount with which the estimated contained amount can be assumed to have not reached the upper limit contained amount yet. The lower limit determination amount is less than the upper limit contained amount.
[0074] When the control unit **40** determines that the error condition is satisfied, the processing proceeds to step S**18**. When the control unit **40** determines that the error condition is not satisfied, the processing proceeds to step S**19**.
[0075] In step S**18**, the control unit **40** executes error notification processing. In this processing, the control unit **40** makes the notification unit **42** issue an error notification related to the waste liquid.

As described above, the control unit **40** makes the notification unit **42** issue an error notification related to the waste liquid contained in the waste liquid container **51**, based on a result of the detection by the liquid amount detection unit **37** and a result of the estimation in step S**12**. Specifically, the control unit **40** makes the notification unit **42** issue a notification related to the contained amount of the waste liquid contained in the waste liquid container **51**, based on a result of the estimation in step S**12**. This error notification processing corresponds to an example of notification processing.

[0076] In step S**19**, the control unit **40** determines whether the estimated contained amount is equal to or more than a prescribed contained amount. The prescribed contained amount is a threshold of the contained amount less than the upper limit contained amount. The prescribed contained amount is a threshold indicating that the contained amount almost reaches the upper limit contained amount, meaning that the waste liquid container **51** for replacement may be prepared.

[0077] Upon determining that the estimated contained amount is less than the prescribed contained amount, the control unit **40** terminates the waste liquid control processing. When the control unit **40** determines that the estimated contained amount is equal to or more than the prescribed contained amount, the processing proceeds to step S**20**.

[0078] In step S**20**, the control unit **40** determines whether the upper limit contained amount is detected as the contained amount of the waste liquid contained in the waste liquid container **51** based on the result of the detection by the liquid amount detection unit **37**. Specifically, the control unit **40** determines that the contained amount of the waste liquid contained in the waste liquid container **51** has reached the threshold based on the result of the detection by the liquid amount detection unit **37**. Such processing corresponds to an example of determination processing.

[0079] When the control unit **40** determines that the upper limit contained amount is detected, the processing proceeds to step S**21**. When the control unit **40** determines that the upper limit contained amount is not detected, the processing proceeds to step S**22**.

[0080] In step S**21**, the control unit **40** executes upper limit contained amount notification processing. In this processing, the control unit **40** makes the notification unit **42** issue a notification indicating that the upper limit contained amount has been reached. Thus, the control unit **40** makes the notification unit **42** issue the notification indicating that the contained amount of the waste liquid contained in the waste liquid container **51** has reached the upper limit contained amount, based on the result of the detection by the liquid amount detection unit **37**. Specifically, the control unit **40** makes the notification unit **42** issue a notification related to the contained amount of the waste liquid contained in the waste liquid container **51**, based on the result of the detection by the liquid amount detection unit **37**. Such upper limit contained amount notification processing corresponds to an example of notification processing.

[0081] In step S**22**, the control unit **40** executes prescribed contained amount notification processing. In this processing, the control unit **40** makes the notification unit **42** issue a notification indicating that the upper limit contained amount has not been reached yet but the prescribed contained amount has been reached. Thus, the control unit **40** makes the notification unit **42** issue a notification indicating that the contained amount of the waste liquid contained in the waste liquid container **51** has reached the prescribed contained amount based on a result of estimation in step S**12**. Specifically, the control unit **40** makes the notification unit **42** issue a notification related to the contained amount of the waste liquid contained in the waste liquid container **51**, based on a result of the estimation in step S**12**. Such prescribed contained amount notification processing corresponds to an example of notification processing.

[0082] The control unit **40** which initializes the cumulative contained amount and the estimated contained amount when the waste liquid container **51** is attached to the waste liquid attachment portion **35**, may alternatively initialize the cumulative contained amount and the estimated contained amount when the waste liquid container **51** is detached from the waste liquid attachment portion **35**.

Transition of Estimated Contained Amount

[0083] Transition of the estimated contained amount is described with reference to FIG. **5**. Specifically, the relationship between the estimated contained amount and time will be described as transition of the estimated contained amount. In FIG. **5**, a graph **70** is a graph illustrating a comparative example indicating the estimated contained amount estimated without considering the evaporation amount of the waste liquid. That is, the graph **70** is a graph corresponding to the calculation result of the cumulative contained amount. A graph **71** is a graph illustrating an example indicating an estimated contained amount estimated in consideration of the evaporation amount of the waste liquid.

[0084] As illustrated in FIG. **5**, according to the graph **70** illustrating the comparative example, the waste liquid container **51** is attached to the waste liquid attachment portion **35** at a timing indicated by a reference numeral T**10**. As a result of adding the waste liquid to the waste liquid container **51** at a timing indicated by a reference numeral T**11**, the estimated contained amount of the waste liquid is calculated as a contained amount C**13**. Then, as a result of further adding the waste liquid to the waste liquid container **51** at a timing indicated by a reference numeral T**12**, the estimated contained amount of the waste liquid is calculated as a contained amount C**16**. Thereafter, as a result of the waste liquid being further added to the waste liquid container **51** at the timing indicated by a reference numeral T**13**, the estimated contained amount of the waste liquid is calculated as a contained amount C**17**.

[0085] In the graph **71** illustrating the example, the estimated contained amount of the waste liquid is calculated as a contained amount C**11** at the timing indicated by the reference numeral T**12** as a result of subtraction of the estimated contained amount of the waste liquid due to evaporation of the waste liquid after the timing indicated by the reference numeral T**11**. As a result of further adding the waste liquid to the waste liquid container **51** at a timing indicated by a reference numeral T**12**, the estimated contained amount of the waste liquid is calculated as a contained amount C**14**.

[0086] The estimated contained amount of the waste liquid is calculated as a contained amount C**12** at a timing indicated by a reference numeral T**13** as a result of subtraction of the estimated contained amount of the waste liquid due to evaporation of the waste liquid after the timing indicated by the reference numeral T**12**.

[0087] After the timing indicated by the reference numeral T**13**, the estimated contained amount becomes less than the estimated lower limit amount. In this case, the waste liquid is not evaporated, and the estimated contained amount of the waste liquid is maintained at the contained amount C**12**. As a result of further adding the waste liquid to the waste liquid container **51** at a timing indicated by a reference numeral T**14**, the estimated contained amount of the waste liquid is calculated as a contained amount C**15**.

[0088] As described above, according to the example in which the evaporation of the waste liquid is considered, the estimation accuracy of the estimated contained amount can be improved compared with the comparative example in which the evaporation of the waste liquid is not considered. Specifically, in a case where the contained amount C**17** is the upper limit contained amount, the estimated contained amount reaches the upper limit contained amount in the comparative example, and there is a need to replace the waste liquid container **51** which is in a state of being capable of containing the waste liquid. On the other hand, in the example, the estimated contained amount does not reach the upper limit contained amount, and there is no need to replace the waste liquid container **51** which is in a state of being capable of containing the waste liquid.

Operations and Effects of First Embodiment

[0089] Operations and effects of the first embodiment are described below.

[0090] (1-1) The liquid amount detection unit **37** detects that the contained amount of the waste liquid contained in the waste liquid container **51** has reached the upper limit contained amount. The control unit **40** calculates the evaporation amount of the waste liquid by which the waste liquid contained in the waste liquid container **51** evaporates through the atmosphere opening hole **56**

provided in the waste liquid container **51**. The control unit **40** estimates the contained amount of the waste liquid contained in the waste liquid container **51** based on the additional amount of the waste liquid added to the waste liquid container **51** and the calculation result. According to this configuration, in a case where the waste liquid container **51** provided with the atmosphere opening hole **56** is used, even if the contained amount of the waste liquid contained in the waste liquid container **51** decreases due to evaporation of the waste liquid, it can be determined that the contained amount of the waste liquid has reached the threshold based on the result of the detection by the liquid amount detection unit **37**. Furthermore, the contained amount of the waste liquid contained in the waste liquid container **51** can be estimated based on the additional amount of the waste liquid and the evaporation amount of the waste liquid. Thus, the accuracy of the estimation for the contained amount of the waste liquid can be improved in consideration of the evaporation amount of the waste liquid, while improving the reliability of the recognition that the contained amount of the waste liquid has reached the threshold. Thus, improvement in the function related to the contained amount of the waste liquid contained in the waste liquid container **51** can be achieved.
[0091] (1-2) The control unit **40** makes the notification unit **42** issue a notification related to the contained amount of the waste liquid contained in the waste liquid container **51** based on the estimation result. According to this configuration, the notification can be issued related to the contained amount of the waste liquid estimated based on the additional amount of the waste liquid and the evaporation amount of the waste liquid. Accordingly, the user can be provided with information related to the contained amount of the waste liquid with improved estimation accuracy. Thus, improvement in the function related to the contained amount of the waste liquid contained in the waste liquid container **51** can be achieved.
[0092] (1-3) The control unit **40** makes the notification unit **42** issue a notification indicating that the contained amount of the waste liquid contained in the waste liquid container **51** has reached the prescribed contained amount based on a result of estimation. According to this configuration, it is possible to detect that the contained amount of the waste liquid has reached the upper limit contained amount by the liquid amount detection unit **37**, and to issue the notification indicating that the contained amount of the waste liquid has reached the prescribed contained amount that is less than the upper limit contained amount. Accordingly, the contained amount of the waste liquid reaching the prescribed contained amount can be estimated with improved accuracy, while improving the reliability of the recognition that the contained amount of the waste liquid has reached the upper limit contained amount. Thus, improvement in the function related to the contained amount of the waste liquid contained in the waste liquid container **51** can be achieved.
[0093] (1-4) The control unit **40** makes the notification unit **42** issue an error notification related to the waste liquid contained in the waste liquid container **51**, based on a result of the detection by the liquid amount detection unit **37** and a result of the estimation. According to this configuration, the error notification can be issued that is related to the waste liquid contained in the waste liquid container **51**, by using both of the detection result with improved reliability and the estimation result with improved estimation accuracy. Thus, improvement in the function related to the contained amount of the waste liquid contained in the waste liquid container **51** can be achieved.
[0094] (1-5) The waste liquid container **51** is replaceable. According to this configuration, the contained amount of the waste liquid reaching the threshold can be reliably recognized, and the waste liquid container **51** can be replaced in accordance with the contained amount of the waste liquid estimated with improved accuracy. Thus, improvement in the function related to the contained amount of the waste liquid contained in the waste liquid container **51** can be achieved.

Second Embodiment

[0095] Next, a second embodiment is described. In the following description, the overlapping description of the same configurations as those of the above-described embodiment will be omitted or simplified, and the configurations different from those of the above-described embodiment will be described.

Configuration of Waste Liquid Container **51**

[0096] As illustrated in FIG. **6**, in the second embodiment, the waste liquid container case **52** includes a second storage portion **62**. Thus, the waste liquid container **51** includes the second storage portion **62**. The second storage portion **62** is provided in a flow path through which the first storage portion **61** and the atmosphere opening hole **56** communicate. The second storage portion **62** stores the waste liquid. Specifically, the second storage portion **62** includes a second storage chamber capable of storing the waste liquid.

[0097] The second storage portion **62** is capable of storing a predetermined contained amount of the waste liquid. The predetermined contained amount is obtained by subtracting the prescribed contained amount from the upper limit contained amount. The contained amount of the waste liquid contained in the waste liquid containing portion **60**, the first storage portion **61**, and the second storage portion **62** corresponds to the upper limit contained amount of the waste liquid in the waste liquid container **51**.

[0098] The first partition wall **64** is provided between the waste liquid containing portion **60** and the second storage portion **62**. The first partition wall **64** provides a partition between the waste liquid containing portion **60** and the second storage portion **62**. Accordingly, the waste liquid containing portion **60**, the first storage portion **61**, and the atmosphere opening portion **63**, communicate with the second storage portion **62**, and evaporation of the waste liquid contained in the second reservoir **62** is promoted through the atmosphere opening hole **56**.

[0099] The second partition wall **66** is provided between the first storage portion **61** and the second storage portion **62**. The second partition wall **66** provides a partition between the first storage portion **61** and the second storage portion **62**. The second partition wall **66** includes a second communication portion **67**. The second communication portion **67** is provided to establish communication between the first storage portion **61** and the second storage portion **62**. The second communication portion **67** is provided at a position higher than the inner bottom surface **52**B of the waste liquid container case **52**. The second communication portion **67** may have a recessed shape.

[0100] The first storage portion **61** is provided in a flow path through which the waste liquid containing portion **60** and the atmosphere opening hole **56** communicate. Specifically, the first storage portion **61** is provided in a flow path through which the waste liquid containing portion **60** and the second storage portion **62** communicate. The liquid amount detection unit **37** detects that the contained amount of the waste liquid has reached the prescribed contained amount as the threshold. The contained amount of the waste liquid contained in the waste liquid containing portion **60** and the first storage portion **61** corresponds to the prescribed contained amount of the waste liquid in the waste liquid container **51**.

[0101] The waste liquid container case **52** includes a third partition wall **68**. The third partition wall **68** is provided between the second storage portion **62** and the atmosphere opening hole **56**. The third partition wall **68** provides a partition between the second storage portion **62** and the atmosphere opening portion **63**.

[0102] The third partition wall **68** is lower than the outer wall **52**A of the waste liquid container case **52**. Therefore, a gap is provided between the third partition wall **68** and the waste liquid container cover **53**. Accordingly, the second storage portion **62** communicates with the atmosphere opening portion **63**. Thus, the evaporation of the waste liquid contained in the second storage portion **62** is promoted through the atmosphere opening hole **56**.

[0103] The liquid amount detection unit **37** can detect that the waste liquid is or is not stored in the first storage portion **61**, via the reflection portion **55**. Specifically, the liquid amount detection unit **37** can detect that the waste liquid of the prescribed contained amount is or is not contained in the waste liquid container **51** via the reflection portion **55**.

[0104] When the waste liquid is further injected into the waste liquid container **51** after the waste liquid of the prescribed contained amount has been contained in the waste liquid containing portion

60 and the first storage portion **61**, the waste liquid stored in the first storage portion **61** enters the second storage portion **62** via the second communication portion **67**.

[0105] If the waste liquid is further injected into the waste liquid container **51** after the waste liquid of the upper limit contained amount has been contained in the waste liquid containing portion **60**, the first storage portion **61**, and the second storage portion **62**, the waste liquid overflowed from the waste liquid containing portion **60**, the first storage portion **61**, and the second storage portion **62** enters the atmosphere opening portion **63**. Therefore, when the contained amount of the waste liquid contained in the waste liquid container **51** reaches the upper limit contained amount, the waste liquid container **51** needs to be replaced.

Waste Liquid Control Processing

[0106] As illustrated in FIG. **7**, in steps S**12** to S**17** of the waste liquid control processing, the control unit **40** controls the estimated contained amount of the first embodiment as a first estimated contained amount. In step S**12**, the control unit **40** executes first estimation processing in a manner similar to that for the estimation processing in the first embodiment. In step S**13**, as in the first embodiment, the control unit **40** determines whether the first estimated contained amount estimated when the current waste liquid condition is satisfied is less than the estimated lower limit amount. In step S**14**, the control unit **40** executes first estimated contained amount correction processing in a manner that is the same as that for the estimated contained amount correction processing in the first embodiment. In step S**16**, the control unit **40** executes first estimated contained amount addition processing in a manner similar to that for the estimated contained amount addition processing in the first embodiment.

[0107] In step S**17** of the waste liquid control processing, the error condition is satisfied when the first estimated contained amount is equal to or more than the upper limit determination amount but the amount is detected by the liquid amount detection unit **37** to be not equal to or more than the prescribed contained amount. The upper limit determination amount is an amount with which the first estimated contained amount can be assumed to have reached the prescribed contained amount. The upper limit determination amount may be equal to or more than the prescribed contained amount or may be less than the prescribed contained amount. The error condition is satisfied when the first estimated contained amount is equal to or less than a lower limit determination amount but the amount detected by the liquid amount detection unit **37** is equal to or more than the prescribed contained amount. The lower limit determination amount is an amount with which the first estimated contained amount can be assumed to have not reached the prescribed contained amount yet. The lower limit determination amount is less than the prescribed contained amount. When the control unit **40** determines that the error condition is not satisfied, the processing proceeds to step S**30**.

[0108] In step S**30**, the control unit **40** determines whether the contained amount of the waste liquid contained in the waste liquid container **51** is detected to be equal to or more than the prescribed contained amount, based on the result of the detection of the liquid amount detection unit **37**. Upon determining that the contained amount of the waste liquid being equal to or more than the prescribed contained amount is not detected, the control unit **40** terminates the waste liquid control processing. When the control unit **40** determines that the contained amount of the waste liquid is detected to be equal to or more than the prescribed contained amount, the processing proceeds to step S**31**.

[0109] In step S**31**, the control unit **40** executes second estimation processing. In this processing, the control unit **40** reads, from the memory, a second estimated contained amount of the waste liquid estimated at the point of the previous satisfaction of the waste liquid condition. The second estimated contained amount is a contained amount of waste liquid obtained as a sum of the prescribed contained amount and the contained amount of the waste liquid added to the waste liquid container **51** after the contained amount of the waste liquid has been detected to be equal to or more than the prescribed contained amount. The control unit **40** subtracts the evaporation amount of the waste liquid from the previous second estimated contained amount of the waste liquid. Thus, the control unit **40** estimates the current second estimated contained amount not including the additional amount of the waste liquid. In this way, the control unit **40** estimates the contained amount of the waste liquid contained in the waste liquid container **51** based on the calculation result.

[0110] In step S**32**, the control unit **40** executes second estimated contained amount addition processing. In this processing, the control unit **40** records a result obtained by adding the calculated additional amount of the waste liquid to the second estimated contained amount in the memory as the current second estimated contained amount.

[0111] In this way, the control unit **40** estimates the contained amount of the waste liquid contained in the waste liquid container **51** based on the calculation result and the additional amount of the waste liquid. The second estimation processing and the estimated contained amount addition processing correspond to an example of estimation processing.

[0112] In step S**33**, the control unit **40** determines whether the second estimated contained amount is equal to or more than the upper limit contained amount. When the control unit **40** determines that the second estimated contained amount is less than the upper limit contained amount, the processing proceeds to step S**22**. When the control unit **40** determines that the second estimated contained amount is equal to or more than the upper limit contained amount, the processing proceeds to step S**21**.

[0113] Thus, in step S**22**, the control unit **40** makes the notification unit **42** issue the notification indicating that the contained amount of the waste liquid contained in the waste liquid container **51** has reached the prescribed contained amount, based on the result of the detection by the liquid amount detection unit **37**. Specifically, the control unit **40** makes the notification unit **42** issue a notification related to the contained amount of the waste liquid contained in the waste liquid container **51**, based on the result of the detection by the liquid amount detection unit **37**. Such prescribed contained amount notification processing corresponds to an example of notification processing.

[0114] In step S**21**, after the contained amount of the waste liquid contained in the waste liquid container **51** has reached the prescribed contained amount, the control unit **40** makes the notification unit **42** issue a notification indicating that the contained amount of the waste liquid contained in the waste liquid container **51** has reached the upper limit contained amount based on the result of estimation in step S**31**. Specifically, the control unit **40** makes the notification unit **42** issue a notification related to the contained amount of the waste liquid contained in the waste liquid container **51**, based on a result of the estimation in step S**31**. Such prescribed contained amount notification processing corresponds to an example of notification processing.

[0115] The control unit **40** which initializes the second estimated contained amount when the waste liquid container **51** is attached to the waste liquid attachment portion **35**, may alternatively initialize second estimated contained amount when the waste liquid container **51** is detached from the waste liquid attachment portion **35**.

Operations and Effects of Second Embodiment

[0116] Operations and effects of the second embodiment are described below.

[0117] (2-1) The liquid amount detection unit **37** can detect that the contained amount of the waste liquid has reached the prescribed contained amount, and a notification indicating that the contained amount of the waste liquid has reached the prescribed contained amount and the upper limit contained amount. Accordingly, the contained amount of the waste liquid reaching the upper limit contained amount can be estimated with improved accuracy, while improving the reliability of the recognition that the contained amount of the waste liquid has reached the prescribed contained amount. Thus, improvement in the function related to the contained amount of the waste liquid contained in the waste liquid container **51** can be achieved.

Modifications

[0118] The embodiments may be modified as follows for implementation. The embodiments and modifications described below may be combined for implementation insofar as they are not technically inconsistent. [0119] The control unit **40** may execute the waste liquid control processing before the liquid is ejected from the liquid ejection unit **11** as the waste liquid, or may execute the waste liquid control processing after the liquid has been ejected from the liquid ejection unit **11** as the waste liquid. [0120] The control unit **40** may select an evaporation rate corresponding to the installation environment of the liquid ejection apparatus **10** from among a plurality of types of evaporation rates. The installation environment may include at least one of temperature and humidity. The liquid ejection apparatus **10** may include an environment detection unit that detects at least one of temperature and humidity. The control unit **40** may select the evaporation rate corresponding to the installation environment of the liquid ejection apparatus **10** as a result of referring to at least one of the installation environment at the time of previous satisfaction of the waste liquid condition and the installation environment at the time of current satisfaction of the waste liquid condition. [0121] The control unit **40** may select an evaporation rate corresponding to the type of the waste liquid container **51** attached to the waste liquid attachment portion **35**, from among a plurality of types of evaporation rates. The control unit **40** may determine the type of the waste liquid container **51** attached to the waste liquid attachment portion **35**, based on the result of detection by the container type detection unit **38**. The control unit **40** may select the evaporation rate corresponding to the shape of the atmosphere opening hole **56** provided in the waste liquid container **51**. The control unit **40** may select an evaporation rate in accordance with the presence or absence of the atmosphere opening hole **56** in the waste liquid container **51**. When the waste liquid container **51** provided with the atmosphere opening hole **56** is attached, the control unit **40** may select an evaporation rate higher than that in a case where the waste liquid container **51** not provided with the atmosphere opening hole **56** is attached. [0122] The waste liquid condition may be satisfied when flushing is performed. The control unit **40** may select the additional amount of the waste liquid to be different between a case where the waste liquid condition is satisfied by cleaning and a case where the waste liquid condition is satisfied by flushing. [0123] The liquid ejection apparatus **10** may include an additional amount detection unit that detects the additional amount of the waste liquid led out to the waste liquid container **51**. Accordingly, the control unit **40** may acquire the additional amount of the waste liquid based on the detection result by the additional amount detection unit. [0124] The liquid amount detection unit **37** is configured to include one prism, but is not limited thereto, and may include, for example, two or more prisms. [0125] The liquid amount detection unit **37** may not be an optical sensor. The liquid amount detection unit **37** may detect the contained amount of the waste liquid contained in the waste liquid container **51** based on energization between electrodes. The liquid amount detection unit **37** may detect the contained amount of the waste liquid contained in the waste liquid container **51** based on the vibration waveform of the liquid detected by a piezoelectric sensor. [0126] The liquid ejection apparatus **10** may include the waste liquid container **51**. In this case, the waste liquid container **51** may or may not be detachable from the waste liquid attachment portion **35** of the liquid ejection apparatus **10**. [0127] The medium **90** may be paper, a resin film or a sheet, a composite film of a resin and a metal, a laminate film, a woven fabric, a nonwoven fabric, a metal foil, a metal film, a ceramic sheet, clothing, or the like. [0128] Liquid can be arbitrarily selected as long as it can be recorded on the medium **90** by adhering to the medium **90**. For example, ink includes various compositions such as an aqueous ink, an oil-based ink, a gel ink, a hot melt ink, or the like, including particles of a functional material made of a solid such as pigments or metal particles dissolved, dispersed or mixed in a solvent. [0129] The liquid ejection apparatus **10** is not limited to an inkjet printer, and may be a dot impact printer.

[0130] The expression “at least one of” used in the specification means one or more of desired options. As an example, when the number of options is two, the expression “at least one of” used in the specification means only one option or both of the two options. As another example, when the number of options is three or more, the expression "at least one of" used in the specification means only one option or any combination of two or more options.

Supplementary Note

[0131] The following is a description of the technical ideas and their effects that can be grasped from the above-described embodiments and modifications. This technical concept and its effects can be combined with each other to the extent that they are not technically inconsistent.

[0132] (A) A liquid ejection apparatus includes: a liquid ejection unit configured to eject liquid, a liquid amount detection unit configured to detect, in a waste liquid container capable configured to contain liquid ejected by the liquid ejection unit as waste liquid, that a contained amount of the waste liquid contained in the waste liquid container reaches a threshold, and a control unit, the control unit being configured to execute calculation processing of calculating an evaporation amount of waste liquid when waste liquid contained in the waste liquid container evaporates from the waste liquid container provided with an atmosphere opening hole, and estimation processing of estimating a contained amount of waste liquid contained in the waste liquid container, based on an additional amount of waste liquid when liquid ejected by the liquid ejection unit is added to the waste liquid container as waste liquid and a calculation result in the calculation processing.

[0133] According to this configuration, in a case where the waste liquid container provided with the atmosphere opening hole is used, even if the contained amount of the waste liquid contained in the waste liquid container decreases due to evaporation of the waste liquid, it can be determined that the contained amount of the waste liquid has reached the threshold based on the result of the detection by the liquid amount detection unit. Furthermore, the contained amount of the waste liquid contained in the waste liquid container can be estimated based on the additional amount of the waste liquid and the evaporation amount of the waste liquid. Thus, the accuracy of the estimation for the contained amount of the waste liquid can be improved in consideration of the evaporation amount of the waste liquid, while improving the reliability of the recognition that the contained amount of the waste liquid has reached the threshold. Thus, improvement in the function related to the contained amount of the waste liquid contained in the waste liquid container can be achieved.

[0134] (B) In the liquid ejection apparatus, the control unit may execute notification processing of causing a notification unit to issue a notification related to a contained amount of waste liquid contained in the waste liquid container, based on an estimation result in the estimation processing.

[0135] According to this configuration, the notification can be issued related to the contained amount of the waste liquid estimated based on the additional amount of the waste liquid and the evaporation amount of the waste liquid. Accordingly, the user can be provided with information related to the contained amount of the waste liquid with improved estimation accuracy. Thus, improvement in the function related to the contained amount of the waste liquid contained in the waste liquid container can be achieved.

[0136] (C) In the liquid ejection apparatus, the liquid amount detection unit may detect that a contained amount of waste liquid contained in the waste liquid container reaches an upper limit contained amount as the threshold, and the control unit may execute the notification processing of causing the notification unit to issue a notification indicating that a contained amount of waste liquid contained in the waste liquid container reaches a prescribed contained amount less than the upper limit contained amount, based on an estimation result in the estimation processing.

[0137] According to this configuration, it is possible to detect that the contained amount of the waste liquid has reached the upper limit contained amount by the liquid amount detection unit, and to issue the notification indicating that the contained amount of the waste liquid has reached the prescribed contained amount that is less than the upper limit contained amount. Accordingly, the contained amount of the waste liquid reaching the prescribed contained amount can be estimated with improved accuracy, while improving the reliability of the recognition that the contained amount of the waste liquid has reached the upper limit contained amount. Thus, improvement in the function related to the contained amount of the waste liquid contained in the waste liquid container can be achieved.

[0138] (D) In the liquid ejection apparatus, the waste liquid container may include: a waste liquid containing portion configured to contain liquid ejected from the liquid ejection unit as waste liquid, a first storage portion configured to store waste liquid in a flow path communicating the waste liquid containing portion and the atmosphere opening hole, and a reflection portion at which a reflection state of light is changed by waste liquid stored in the first storage portion, the liquid amount detection unit may be an optical sensor, the optical sensor may include a light emitting portion configured to emit light toward the reflection portion and a light receiving portion configured to receive the light reflected by the reflection portion, and the optical sensor may detect that a contained amount of waste liquid contained in the waste liquid container reaches the upper limit contained amount, based on light received by the light receiving portion. According to this configuration, the same effects as in (C) can be exhibited.

[0139] (E) In the liquid ejection apparatus, the liquid amount detection unit may detect that a contained amount of waste liquid contained in the waste liquid container reaches a prescribed contained amount as the threshold that is less than an upper limit contained amount, and the control unit may execute the notification processing of causing the notification unit to issue a notification indicating that a contained amount of waste liquid contained in the waste liquid container reaches the prescribed contained amount, based on a detection result by the liquid amount detection unit, and after a contained amount of waste liquid contained in the waste liquid container reaches the prescribed contained amount, causing the notification unit to issue a notification indicating that a contained amount of waste liquid contained in the waste liquid container reaches the upper limit contained amount, based on an estimation result in the estimation processing.

[0140] According to this configuration, the liquid amount detection unit can detect that the contained amount of the waste liquid has reached the prescribed contained amount less than the upper limit contained amount, and the notification indicating that the contained amount of the waste liquid has reached the prescribed contained amount and the upper limit contained amount can be issued. Accordingly, the contained amount of the waste liquid reaching the upper limit contained amount can be estimated with improved accuracy, while improving the reliability of the recognition that the contained amount of the waste liquid has reached the prescribed contained amount. Thus, improvement in the function related to the contained amount of the waste liquid contained in the waste liquid container can be achieved.

[0141] (F) In the liquid ejection apparatus, the waste liquid container may include: a waste liquid containing portion configured to contain liquid ejected from the liquid ejection unit as waste liquid, a first storage portion configured to store the waste liquid in a flow path communicating the waste liquid containing portion and the atmosphere opening hole, a second storage portion configured to store waste liquid in a flow path communicating the first storage portion and the atmosphere opening hole, and a reflection portion at which a reflection state of light is changed by waste liquid stored in the first storage portion, the second storage portion may be configured to store a predetermined contained amount of waste liquid, the predetermined contained amount may be obtained by subtracting the prescribed contained amount from the upper limit contained amount, the liquid amount detection unit may be an optical sensor, the optical sensor may include a light emitting portion configured to emit light toward the reflection portion and a light receiving portion configured to receive light reflected by the reflection portion, and the optical sensor may detect that a contained amount of waste liquid contained in the waste liquid container reaches the prescribed contained amount, based on the light received by the light receiving portion. According to this configuration, the same effects as in (E) can be exhibited.

[0142] (G) In the liquid ejection apparatus, the control unit may execute the notification processing of causing the notification unit to issue an error notification related to waste liquid contained in the waste liquid container, based on a detection result by the liquid amount detection unit and an

estimation result in the estimation processing.
[0143] According to this configuration, the error notification can be issued that is related to the waste liquid contained in the waste liquid container, by using both of the detection result with improved reliability and the estimation result with improved estimation accuracy. Thus, improvement in the function related to the contained amount of the waste liquid contained in the waste liquid container can be achieved.
[0144] (H) In the liquid ejection apparatus, the waste liquid container may be replaceable.
[0145] According to this configuration, the contained amount of the waste liquid reaching the threshold can be reliably recognized, and the waste liquid container can be replaced in accordance with the contained amount of the waste liquid estimated with improved accuracy. Thus, improvement in the function related to the contained amount of the waste liquid contained in the waste liquid container can be achieved.
[0146] (I) The liquid ejection apparatus may include the waste liquid container configured to contain liquid ejected by the liquid ejection unit as waste liquid, and the waste liquid container may include the atmosphere opening hole configured to evaporate waste liquid contained in the waste liquid container therethrough. According to this configuration, the same effects as in (A) can be exhibited.
[0147] (J) A control method for a liquid ejection apparatus configured to perform control related to a contained amount of waste liquid that is liquid ejected by a liquid ejection unit and contained in a waste liquid container as waste liquid, the control method including executing: determination processing of determining that a contained amount of waste liquid contained in the waste liquid container reaches a threshold, based on a detection result by a liquid amount detection unit, calculation processing of calculating an evaporation amount of waste liquid when waste liquid contained in the waste liquid container evaporates from the waste liquid container provided with an atmosphere opening hole, and estimation processing of estimating a contained amount of waste liquid contained in the waste liquid container, based on an additional amount of waste liquid when liquid ejected by the liquid ejection unit is added to the waste liquid container as waste liquid and a calculation result in the calculation processing. According to this configuration, the same effects as in (A) can be exhibited.

Claims

1. A liquid ejection apparatus comprising: a liquid ejection unit configured to eject liquid; a liquid amount detection unit configured to detect, in a waste liquid container configured to contain liquid ejected by the liquid ejection unit as waste liquid, that a contained amount of waste liquid contained in the waste liquid container reaches a threshold; and a control unit, the control unit being configured to execute: calculation processing of calculating an evaporation amount of waste liquid when waste liquid contained in the waste liquid container evaporates from the waste liquid container provided with an atmosphere opening hole; and estimation processing of estimating a contained amount of waste liquid contained in the waste liquid container, based on an additional amount of waste liquid when liquid ejected by the liquid ejection unit is added to the waste liquid container as waste liquid and a calculation result in the calculation processing.
2. The liquid ejection apparatus according to claim 1, wherein the control unit executes notification processing of causing a notification unit to issue a notification related to a contained amount of waste liquid contained in the waste liquid container, based on an estimation result in the estimation processing.
3. The liquid ejection apparatus according to claim 2, wherein the liquid amount detection unit detects that a contained amount of waste liquid contained in the waste liquid container reaches an upper limit contained amount as the threshold, and the control unit executes the notification processing of causing the notification unit to issue a notification indicating that a contained amount

of waste liquid contained in the waste liquid container reaches a prescribed contained amount less than the upper limit contained amount, based on an estimation result in the estimation processing.

4. The liquid ejection apparatus according to claim 3, wherein the waste liquid container includes: a waste liquid containing portion configured to contain liquid ejected from the liquid ejection unit as waste liquid; a first storage portion configured to store waste liquid in a flow path communicating the waste liquid containing portion and the atmosphere opening hole; and a reflection portion at which a reflection state of light is changed by waste liquid stored in the first storage portion, the liquid amount detection unit is an optical sensor, the optical sensor includes a light emitting portion configured to emit light toward the reflection portion and a light receiving portion configured to receive light reflected by the reflection portion, and the optical sensor detects that a contained amount of waste liquid contained in the waste liquid container reaches the upper limit contained amount, based on light received by the light receiving portion.

5. The liquid ejection apparatus according to claim 2, wherein the liquid amount detection unit detects that a contained amount of waste liquid contained in the waste liquid container reaches a prescribed contained amount as the threshold that is less than an upper limit contained amount, and the control unit executes the notification processing of causing the notification unit to issue a notification indicating that a contained amount of waste liquid contained in the waste liquid container reaches the prescribed contained amount, based on a detection result by the liquid amount detection unit, and after a contained amount of waste liquid contained in the waste liquid container reaches the prescribed contained amount, causing the notification unit to issue a notification indicating that a contained amount of waste liquid contained in the waste liquid container reaches the upper limit contained amount, based on an estimation result in the estimation processing.

6. The liquid ejection apparatus according to claim 5, wherein the waste liquid container includes: a waste liquid containing portion configured to contain liquid ejected from the liquid ejection unit as waste liquid; a first storage portion configured to store the waste liquid in a flow path communicating the waste liquid containing portion and the atmosphere opening hole; a second storage portion configured to store waste liquid in a flow path communicating the first storage portion and the atmosphere opening hole; and a reflection portion at which a reflection state of light is changed by waste liquid stored in the first storage portion, the second storage portion is configured to store a predetermined contained amount of waste liquid, the predetermined contained amount is obtained by subtracting the prescribed contained amount from the upper limit contained amount, the liquid amount detection unit is an optical sensor, the optical sensor includes a light emitting portion configured to emit light toward the reflection portion, and a light receiving portion configured to receive light reflected by the reflection portion, and the optical sensor detects that a contained amount of waste liquid contained in the waste liquid container reaches the prescribed contained amount, based on the light received by the light receiving portion.

7. The liquid ejection apparatus according to claim 2, wherein the control unit executes the notification processing of causing the notification unit to issue an error notification related to waste liquid contained in the waste liquid container, based on a detection result by the liquid amount detection unit and an estimation result in the estimation processing.

8. The liquid ejection apparatus according to claim 1, wherein the waste liquid container is replaceable.

9. The liquid ejection apparatus according to claim 1, comprising the waste liquid container configured to contain liquid ejected by the liquid ejection unit as waste liquid, wherein the waste liquid container includes the atmosphere opening hole configured to evaporate waste liquid contained in the waste liquid container therethrough.

10. A control method for a liquid ejection apparatus configured to perform control related to a contained amount of waste liquid that is liquid ejected by a liquid ejection unit and contained in a waste liquid container as waste liquid, the control method comprising executing: determination processing of determining that a contained amount of waste liquid contained in the waste liquid container reaches a threshold, based on a detection result by a liquid amount detection unit; calculation processing of calculating an evaporation amount of waste liquid when waste liquid contained in the waste liquid container evaporates from the waste liquid container provided with an atmosphere opening hole; and estimation processing of estimating a contained amount of waste liquid contained in the waste liquid container, based on an additional amount of waste liquid when liquid ejected by the liquid ejection unit is added to the waste liquid container as waste liquid and a calculation result in the calculation processing.